

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. MTB-504: Differential Geometry

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Give the definition of a regular curve in Euclidean 3-space. Show that a curve $\alpha : \mathbb{R} \rightarrow \mathbb{R}^3$ defined by $\alpha(t) = (t^3, t^2, 0)$ for $t \in \mathbb{R}$ is not regular. [2]
- (b) Give the definition of normal plane and rectifying plane of a regular curve in $\mathbb{R}^3$. [2]
- (c) Give the definition of Frenet-Serret apparatus of a unit speed curve $\alpha : (a, b) \rightarrow \mathbb{R}^3$. [2]
- (d) State the local fundamental theorem for regular curves in Euclidean 3-space. [2]
- (e) Give the definitions of the first fundamental form and second fundamental form for a simple surface in Euclidean 3-space. [2]
- (f) Give the definitions of Christoffel symbols of the first kind and Christoffel symbols of the second kind for a simple surface in Euclidean 3-space. [2]
- (g) Give the definitions of Gaussian curvature and the mean curvature for a simple surface in Euclidean 3-space. [2]

Question 2.

- (a) State and prove the Frenet-Serret formulas for a unit speed curve in Euclidean 3-space. [7]
- (b) Give the definition of a general helix in Euclidean 3-space. Prove that a unit speed curve $\alpha(s)$ with $\kappa(s) \neq 0$ for every s is a general helix if and only if $\tau = c\kappa$ for some real constant c . [7]

Question 3.

- (a) Give the definition of the osculating plane of a regular curve in Euclidean 3-space. For a unit speed curve $\alpha : (a, b) \rightarrow \mathbb{R}^3$, if $\kappa > 0$ and $\tau \neq 0$, then show that the equation of the osculating plane at a point P is:

$$[R - \alpha, \alpha', \alpha''] = 0$$

where symbols have their usual meanings. [7]

- (b) Let α be a unit speed curve which lies on a sphere of radius $a > 0$ centered at the point x_0 . If $\tau \neq 0$, then show that:

$$\alpha(s) - x_0 = -\rho \mathbf{n} - \rho' \sigma \mathbf{b}$$

where $\rho = \frac{1}{\kappa}$ and $\sigma = \frac{1}{\tau}$, and also show that:

$$\rho^2 + (\rho'\sigma)^2 = a^2$$

Do the normal planes of a spherical curve pass through the center? [7]

Question 4.

(a) State and prove the fundamental theorem of curves in Euclidean 3-space. [7]

(b) Let $\alpha(s)$ be a unit speed curve with $\kappa \neq 0$. Show that the following three statements are equivalent:

- (i) $\alpha(s)$ is a planar curve,
- (ii) the binormal vector $\mathbf{b}(s)$ is a constant vector, and
- (iii) the torsion $\tau(s) = 0$ for all s .

[7]

Question 5.

(a) Give the definition of a simple surface of class C^k . Prove that $\mathbf{x} : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by

$$\mathbf{x}(u, v) = \left(\frac{2au}{1+u^2+v^2}, \frac{2av}{1+u^2+v^2}, \frac{u^2+v^2-1}{1+u^2+v^2}a \right)$$

is a simple surface which represents a part of a sphere. [7]

(b) Give the definition of the tangent plane at a point of a simple surface. Let

$$\mathbf{x}(u, v) = (u, v, \sqrt{1-u^2-v^2})$$

be a simple surface. Obtain the tangent plane at the point $\mathbf{x}(1/2, 1/2)$. [7]

Question 6.

(a) Give the definition of the area $S(D)$ of a region D covered by a surface patch $\mathbf{x} : U \rightarrow \mathbb{R}^3$. Obtain the area of a sphere of radius $a > 0$, given by

$$\mathbf{x}(u, v) = (a \sin u \cos v, a \sin u \sin v, a \cos u)$$

[7]

(b) Obtain the Christoffel symbols of the second kind for the simple surface $\mathbf{x}(u, v) = (u, v, uv)$. [7]

Question 7.

(a) State and prove the Gauss and Weingarten formulas for a simple surface $\mathbf{x} : U \rightarrow \mathbb{R}^3$ of class $\geq C^2$. [7]

(b) Obtain the Gaussian curvature of a sphere of radius $a > 0$ in Euclidean 3-space. [7]

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